

Waste to energy

Viability and potential for future development

Davide Simeone

Filip Nyzell

Marija Mihajlovic



LUNDS
UNIVERSITET

Abstract

This paper explores the concept of waste-to-energy (WtE) as a crucial approach to manage residual waste for energy production, focusing on its environmental, technical, and economic dimensions. The study examines WtE plants' operational processes, using Sysav's WtE plant in Malmo as an example. The CO_2 treatment methods and the energy products generated by WtE, such as electricity, heat, and steam, are discussed to highlight their potential benefits.

The economic perspective is analysed across developed countries (Sweden, France, USA) and a developing country (Ghana). In Ghana, the challenges of solid waste management, including insufficient infrastructure, financial strain, and societal reluctance, contribute to a significant economic toll. Conversely, Sweden emerges as a global exemplar in responsible waste management, generating revenue of 100 million USD annually with less than 1% landfill waste.

The economic landscapes of France and the USA reveal a complex interplay between waste-to-energy, environmental concerns, and economic viability. France, with an annual revenue exceeding 2.5 billion USD, faces economic challenges due to varying waste compositions and regulatory compliance. In the USA, where waste-to-energy generates 7.4 billion USD annually, regional variations in waste management policies and technological approaches contribute to a multifaceted economic perspective.

Alternative waste treatments, such as landfills and waste incineration, are compared, emphasising their environmental impact and economic implications. The paper also explores alternative energy production methods, such as coal, solar, wind, and hydroelectric power plants, providing insights into their costs and feasibility.

The conclusion is that waste to energy could bring benefits for the environment and it is economically viable but in some countries the lack of infrastructure makes it difficult and the investment in other power production is cheaper.

Table of contents:

Abstract	1
1. Introduction	3
2. Methods and Results	3
2.1 Method	3
2.2 Waste hierarchy	4
2.3 Power plant	4
2.3.1 CO ₂ treatment	5
2.3.2 Energy products	6
2.4 Alternative waste treatments	6
2.4.1 Landfills	6
2.4.2 Waste incineration	7
2.5 Economy	8
2.5.1 Ghana	8
2.5.2 Sweden	9
2.5.3 France	9
2.5.4 USA	10
2.6 Construction cost	11
3. Discussions and conclusions	13
References	14

1. Introduction

“get away from the caveman approach and look at waste as a resource” said Gary Shepherd, Managing Director of the UK firm Gradsol (<https://www.rts.com>)

Waste contains more than just refuse; it holds valuable resources like energy and materials. Utilising the energy inherent in waste involves the idea of 'waste-to-energy'.

In this discussion, emphasis will be placed solely on residual waste treatment for energy generation, excluding organic waste, which can be processed through methods like gasification, fermentation, or anaerobic digestion.

However, this approach involves burning waste, raising the question: Are the benefits outweighing the drawbacks? A question that will be examined thoroughly in this report.

To begin, one should delve into how waste-to-energy systems operate and compare their environmental impact to conventional waste management methods, such as incineration or landfill disposal.

Subsequently, it's necessary to address the financial aspect: is it economically viable? Is the investment outweighed by better ones, such as wind turbines etc.?

Due to the scope of the paper, limitations have to be made. During the chapter regarding economy, this paper will mainly focus on the developed countries Sweden, France and USA. Sweden because they have a high usage of district heating, France because they have a low usage of district heating in comparison to how much they produce and USA because they have access to a higher amount of capital meaning they have access to high technology. To compare these to a developing country Ghana is chosen.

2. Methods and Results

2.1 Method

Method used in the following report is a literature study of scientific papers, news and company research. First the focus is on evaluating the waste as a resource and the production process of existing Waste to Energy facilities. That is followed up with research and analysis about effects on climate by different methods of waste treatment. A thorough analysis of the economic situation in different countries and their utilisation of WtE was done to get an overview of the world situation. Finally an assessment is made of investment needed for alternative energy production to get a wider perspective on the topic.

2.2 Waste hierarchy

Waste hierarchy is a tool used in management of waste. According to the Waste Framework Directive by the European Union, waste hierarchy is made of 5 steps: prevention, preparing for re-use, recycling, other recovery and disposal. This means that priority is to prevent generation of waste, followed by promoting re-use of products and then recycling when the product has reached the end of life. There is waste that cannot be recycled due to toxic substances or mixed materials, these are at the bottom of the waste hierarchy [1]. To dispose of this kind of waste there are only two options: recovery and generation of energy or disposal on landfills. 2021 municipal waste by EU27 was 49% recycling and composting, 26% waste to energy and 23% landfill. However, the situation varies in different countries. Belgium, Germany, Sweden are some of the countries that have almost no landfills and either recycle or use waste to energy. On the other hand there are countries like Greece, Croatia and Cyprus that have largely landfill and recycle a small percentage but have no waste to energy treatment [2]. To reach the goal of the European Green Deal and make Europe the first net zero gas emissions continent, actions need to be taken across the continent. Waste management and waste to energy could play a big role in bridging the gap between the countries and bringing Europe closer to the Green deal goals [3].

2.3 Power plant

Before proceeding with the discussion it's necessary to understand how these power plants work. The waste-to-energy (WtE) plants transform waste into energy through a Rankine cycle. To delve into the details and analyse step by step can be useful to use an example: the Sysav's WtE plant in Malmo shown in figure 1.

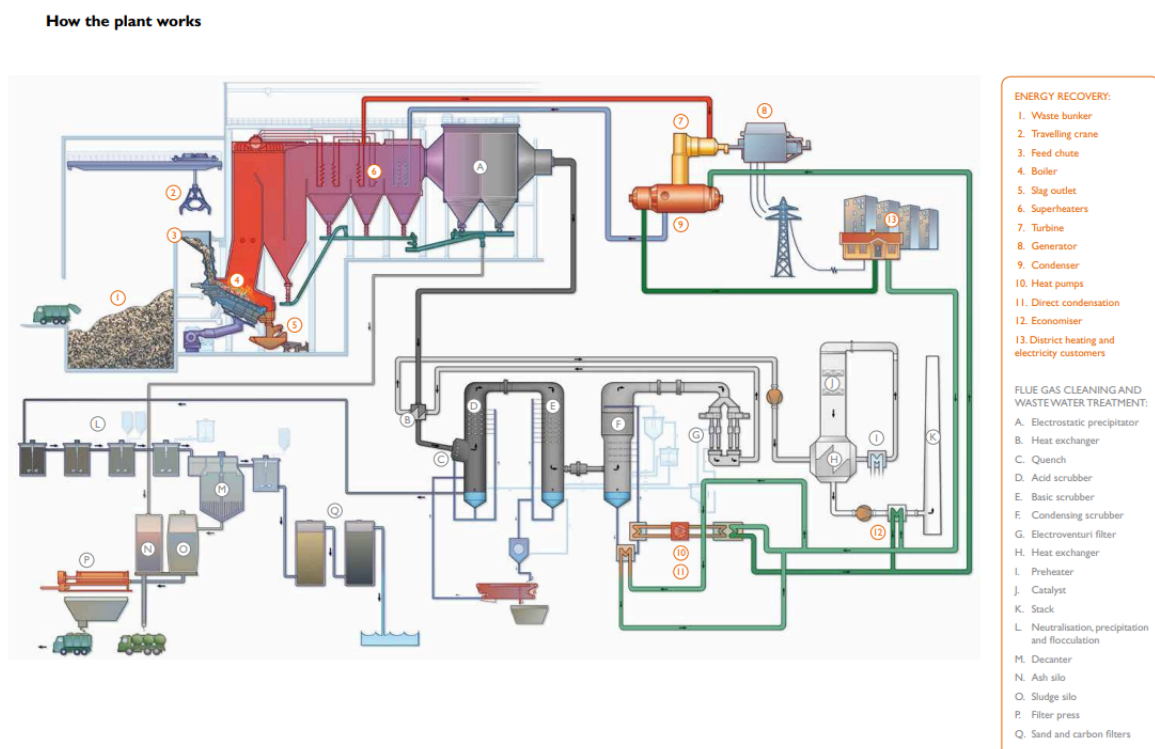


Figure 1: power plant scheme (Heat and electricity from waste Sysav's waste-to-energy plant).[4]

Every year it processes 640 000 tonnes of waste, produces 1 400 000 MWh of district heating and around 250 000 MWh of electricity in total, some of which is used in the plant itself. The WtE plants can have hot-water boilers which can produce only hot water for the district heating network or/and steam boilers which can produce both electricity and hot water.

The heart of the plant is the furnace: its ground is an inclined grate, such that the waste slides down where it is dried, evaporated and combusted. The temperature in the furnace is normally over 1000°C.

During the start-up and shutdown of the plant or when the waste energy level is not high enough to maintain the correct combustion temperature, oil burners are used.

(The Low heating(LHV) value of waste is between 10/15 MJ/kg while oil's LHV is around 40 MJ/kg)

Following combustion the residual materials, collectively referred to as slag, comprise stones, metals, glass, and other inert substances. These remnants are safely channelled into water-filled bunkers for cooling and subsequent recycling or responsible disposal.

The flue gases formed during the combustion have a temperature of at least 850°C. They heat water circulating in tubes. The resulting boiler water reaches its highest temperature of 400°C and a pressure of 40 bar.

The hot high pressure steam can be used to produce electricity, heat or be used as it is in industry processes: in the exemple it is used to produce only electricity and heat: the steam it's channelled in a turbine which drives a generator to produce electricity. After it passes the turbine it is directed to a condenser where its latent heat is harnessed to elevate the temperature of district heating water, which is dispatched through a comprehensive network to provide heat for local homes and facilities. District heating water temperatures typically range from 80 to 115°C.

If the turbine it's not operative or there isn't any, the steam goes directly to the condenser.

Finally when the steam is turned back to water and cooled down, it is pumped again in the system to start the cycle again [4].

2.3.1 CO_2 treatment

This, as most of the power plants which are based on combustion, try to reduce as much CO_2 emissions as possible .

The first line in the treatment of the flue gases is the boiler itself: some of the dust content in the gases stick to the panels walls or to the tubes suspended inside. The cleaning system strikes bars at regular intervals which sets the tubes oscillating. The ash falls down into a dust pocket and is transported to an ash silo.

Then the flue gases pass through an electrostatic precipitator, where the most of the dust contained is captured: the dust particles are given a negative electrical charge when they pass emission

electrodes. Between these emission electrodes are disposed precipitation electrodes, made of metal sheets that are positively charged and attract the dust. At regular intervals they get shaken to remove the captured dust.

When the flue gas leaves the electrostatic filter the ash content is less than 20 mg/m³. From 100 Kg of waste remains only 3 kg of ashes/dust.

Thereafter the gas is subjected to the wet part of the cleaning process: the flue gases pass through three scrubbers, where they undergo a thorough washing process with water sprayed from ceramic nozzles. Diluted in this water, various cleansing chemicals initiate reactions with the different substances present in the flue gases. Periodically, the water is replaced and undergoes a cleaning process.

The penultimate cleaning stage involves a wet electrostatic precipitator, which operates in a similar way to a conventional precipitator, but in a wet environment. In this stage the dust particles are negatively charged and then absorbed and collected by positively charged water spray. Subsequently the water is directed to the water treatment facility.

In the final stage the flue gases pass through a catalyst, where they interact with a ceramic material while ammonia water is injected. During this process nitrogen oxide reacts with the ammonia, reducing it to nitrogen and steam.

Finally the flue gases are released through a 100 metre tall stack [5, 6].

2.3.2 Energy products

In 2018 in Europe, 40 billion kWh electricity and 90 billion kWh heat were generated from Waste-to-Energy plants, using non-recyclable waste. From these, electricity was supplied to 18 million citizens and heat to 15 million. Waste-to-Energy plants also supply steam at suitable pressure and temperature for industrial purposes; it replaces the use of fossil-fuel fired boilers, saving up tonnes of CO₂ emissions per year.

The energy generated by the incineration of 4 tons of garbage is equivalent to 1 ton of oil, 1.6 tons of coal, or 5 tons of wood waste [7].

2.4 Alternative waste treatments

2.4.1 Landfills

A very common way to deal with waste in landfills. It is the oldest way of disposal done by humans. Landfills are sites a bit removed from the city where waste is buried and left to decompose. Procedure is as follows, first a location is chosen and then a hole is dug. The waste is deposited in layers. The first layer is a protective layer. This layer will prevent the liquids from the waste to contaminate ground water under the site. The waste is pressed and compacted before burying to take up less space. At last the waste needs to be covered with wood chips or dirt to prevent animals from digging it up and flies to reproduce and spread disease. During the decomposition the liquid and gas accumulation in the ground needs to be monitored. Some landfills have draining systems

that lead away water. Another useful system is bioreactors such as anaerobic or aerobic bacteria that help and speed up the decomposition of waste in the ground. It can also be mentioned that there are different types of landfills: the municipal ones where people deposit their trash, the industrial one for big companies and the hazardous landfill for toxic waste, also construction and demolition landfill that only accepts waste from buildings. And the difference in waste composition also alters the effects of the landfill [8].

To bring to light some of the negative effects connected to landfill. Following can be mentioned: Municipal waste landfills create toxic gases during decomposition. Three things that contribute to this is bacterial decomposition of organic material, volatilization which means transition of liquid or solid to vapour and lastly the chemical reactions that accrue when different compounds come in contact with each other. Mainly methane and carbon dioxide are released. Methane traps 20 times more solar radiation than carbon dioxide which means that it contributes comparatively more to global warming. Methane stands for 30% of climate change over all and year 2020, 68 million tonnes of methane were released from international landfills [8]. Prognosis for 2025 is that landfills will solely make up 10% of greenhouse gas emissions [9].

Methane can be a more acute danger because it easily spreads fire and can cause explosions. A lot of fires start actually inside the landfill which makes it difficult to detect. This can be dangerous for the environment as well because a fire can damage the protective layer and let the waste contaminate groundwater. Another is that dioxin gas released can be very harmful to the humans in vicinity [9].

Despite the safety measures accidents happen and that can lead to big consequences regarding the environment. Landfills are in fact often responsible for contamination of soil or groundwater. Contaminating materials get in contact with water and compounds like mercury and lead seep into the surrounding area. This is both lethal to animals and plants. Birds are especially vulnerable living near a landfill. A change in their behaviour has been observed. The birds are changing their migration patterns because they can survive on eating waste from landfills during winter [9].

Other areas can also get affected by landfill such as the real estate value of living spaces nearby. Also workers can get hurt and even die on the job due to accidents that still happen around the world on landfills.

2.4.2 Waste incineration

Another way to deal with waste is incineration. This has been viewed as a more green way of managing waste than landfills. Unfortunately during incineration huge amounts of CO_2 are emitted and that makes it a really bad replacement for landfills and can't be considered as carbon neutral or better for climate change in any way. Incineration of 1 Mg of municipal waste is equal to 0.7-1.2 Mg of CO_2 . In some underdeveloped countries the waste is just being incinerated in the open without waste management or flue gas cleansing [10].

There is a rise of projects related to Carbon capture and storage, abbreviated to CCS. In Norway there are around 20 projects like this. One of those projects is led by Statkraft in Trondheim, which is a

Norwegian electricity distributor and producer. In Trondheim there is a WtE facility that will attempt the carbon capture. By doing so they aim to make district heating operation climate neutral by 2040. There is a distinction between biogenic waste and fossil waste. Incineration of biogenic waste is considered carbon neutral due to the existence of a natural cycle for organic matter. Fossil waste does not have such a cycle and therefore emissions from the incineration are considered to contribute to climate change. One third of emissions in Trondheim are fossil. By capturing CO_2 , it would be possible to make incineration of fossil waste carbon neutral, but using the same technology for biogenic waste would also mean that a carbon negative net is possible.

The plan is to mix greenhouse gases with the smoke from the chimney and then use flue gas CO_2 capture technology to extract the CO_2 and then turn it to liquid for easier shipment to intermittent storage. A problem with storage of CO_2 is that it takes a lot of space that is not readily available, a primary calculation showed that they need $7\,000\,m^3$. Therefore the CO_2 needs to be stored in an intermittent storage before being shipped to permanent storage that is under the seabed near Bergen. Estimation is that 21 trucks of 30 tonnes will transport CO_2 from the WtE plant every day [11].

Fortum in Finland launches a pilot programme “Carbon2X” in their incineration plant Riihimäki, where they have found a different way of dealing with the captured carbon. Instead of storage they plan to produce carbon as a raw material to be used in new products. That way waste that usually is incinerated due to inability to sort or recycle can be recycled in a new way [12].

To sum up the treatment of non-recyclable waste with Waste-to-Energy allows the offset of greenhouse gas emissions for several reasons:

- The non-recyclable waste does not end up in landfills anymore.
- The CO_2 produced can be processed with CO_2 capture
- WtE plants recover and supply energy, thus permitting the use of less fossil fuels.
- WtE plants recover metals and minerals, preventing the extraction of further materials.

In particular, the diversion of waste from landfills prevents the production of methane emissions, which is up to 100 times more potent than CO_2 over a 80-year period [13].

2.5 Economy

For the financial aspect of WtE plants the presented order will be as follows: Ghana, Sweden, France and USA. Here the country's revenues of the WtE-plants and technologies will be presented.

2.5.1 Ghana

In Ghana, the escalating challenge of solid waste management, exacerbated by a population surpassing 31.6 million, manifests in an annual waste output of 7.5 million metric tonnes. Shockingly, two-thirds of this waste either ends up in sewers or is incinerated in landfills, resulting in a substantial financial toll. The Accra Metropolitan Assembly expends \$3.45 million annually on waste collection and disposal, while indiscriminate waste disposal costs the country a staggering \$290 million each year.

These concerning statistics underscore the immediate necessity for innovative waste management strategies. WtE solutions emerge as a promising avenue to simultaneously address waste treatment and resource recovery. This study delves into the potential of WtE, estimating that around 4.5 million tons of waste could yield approximately 2 GWh of electricity annually through controlled incineration. Additionally, landfilling could contribute 1.0 – 1.5 GWh of electricity per year, emphasising the critical role WtE can play in transforming waste into a valuable resource.

However, the challenges facing Municipal Solid Waste (MSW) management in Ghana carry direct economic implications. The Ministry of Environment's Waste Management Department is struggling because they have a lack of key staff such as engineering and other key roles, making it hard to make progress. Poor infrastructure makes the economic consequences worse, as bad roads and layout conditions impede waste companies' access to low-income areas, compromising the efficiency of waste management services.

The financial strain intensifies with the reluctance of households and shop owners to pay for waste services, impacting the overall effectiveness of MSW management. Inadequate waste collection and transport vehicles, particularly in rapidly urbanising areas, compound these economic challenges. To sum up, insufficient infrastructure, facilities and the reluctant urban population which further strain logistics, impacting the overall economic investment and risks associated with Waste-to-Energy initiatives [14].

2.5.2 Sweden

Sweden is one of the pioneers regarding waste management. With an annual revenue of \$100 million as of 2022, while also managing to have less than 1% of their waste coming from landfills, their usage of technology is also highly advanced. The usage of Combined Heat and Power(CHP) plants is not uncommon due to their high usage of district heating, resulting in so called incineration with energy recovery. By viewing waste as a resource, Sweden not only minimises environmental impact but also capitalises on economic opportunities through energy regeneration, job creation, innovation, and a sustainable revenue model. This forward-thinking approach has positioned Sweden as a global exemplar in responsible waste management and energy innovation [15].

2.5.3 France

France is a larger country than Sweden, which is shown as their annual revenue from WtE is upwards of \$2.5 billion in 2022. France uses primarily mechanical-grate incineration as their technology. This results in their ability to have three modes for recovering energy. These are electricity, heat and CHP [16, 17].

WtE initiatives have resulted in a complex economic perspective as environmental goals with economic viability clash. From an economic standpoint, France sees waste-to-energy as a potential solution to manage its waste stream while aiming to reduce reliance on landfills and curb greenhouse gas emissions. Waste incineration plants are used to convert waste into energy, contributing to the country's energy mix. However, the economic perspective on waste-to-energy in France faces certain challenges and nuances. One primary economic concern revolves around the balance between incineration costs and energy generation. While waste-to-energy facilities generate electricity and heat, the profitability of these operations heavily depends on factors such as waste composition, technological efficiency, and energy prices. Variations in waste composition can impact the efficiency of energy production and the overall economic viability of the process [17].

Additionally, waste-to-energy facilities in France face concerns from environmental groups concerned about potential air pollution and emissions from incineration. Meeting stringent environmental standards and investing in advanced emission control technologies add to the operational costs, influencing the economic feasibility of these projects. The economic landscape is also influenced by waste management policies and regulations. France has been emphasising waste reduction, recycling, and circular economy strategies, potentially impacting the availability and composition of waste suitable for energy recovery. This dynamic landscape shapes the economic feasibility and long-term sustainability of waste-to-energy projects in the country [16].

Moreover, waste-to-energy initiatives in France contribute to job creation, technology development, and waste infrastructure improvement. However, the economic balance between waste management costs, energy generation, environmental compliance, and changing waste management strategies remains a key consideration in evaluating the overall economic perspective of waste-to-energy in France.

2.5.4 USA

Regarding the USA, their annual revenue of \$7.4 billion makes up to a small but stable amount of produced electricity. The USA mainly uses Mass-burn and Refuse-derived fuel technologies, but these are only for the most advanced part of the country. In other areas (not the north-east and state of Florida) landfills are primarily used.

In the United States, the economic perspective on waste-to-energy presents a multifaceted landscape shaped by varying approaches across different states and regions. The concept of converting waste into energy has gained attention as a potential solution for managing municipal solid waste while addressing energy needs, albeit with diverse economic implications. From an economic standpoint, waste-to-energy initiatives in the US face a blend of opportunities and challenges. On one hand, these projects offer a means of reducing reliance on landfills and contribute to renewable energy generation. Waste incineration plants generate electricity and heat, potentially diversifying the energy mix and aiding in energy security. [18,19]

However, the economic feasibility of these initiatives is influenced by several factors. The economic viability of waste-to-energy endeavours in the US hinges on technology efficiency, waste composition, regulatory compliance, and energy market dynamics. Variations in waste composition across regions affect the energy output and the overall cost-effectiveness of waste-to-energy facilities. Additionally, meeting stringent emission standards and investing in advanced pollution control technologies add to operational costs, impacting the economic feasibility of such projects. [18,19]

Furthermore, waste management policies, recycling efforts, and landfill regulations play a pivotal role in shaping the economic perspective of waste-to-energy. Some states prioritise waste reduction, recycling, and landfill diversion over energy recovery, which can influence the availability and composition of waste suitable for energy conversion. In terms of job creation and economic impact, waste-to-energy initiatives in the US contribute to employment opportunities in facility management, technology development, and waste handling sectors. However, the economic balance between waste management costs, energy production, environmental compliance, and evolving waste management strategies remains a crucial consideration in evaluating the overall economic perspective of waste-to-energy in the US. [18,19]

While waste-to-energy presents potential economic benefits, its feasibility varies across regions due to diverse regulatory frameworks, technological advancements, waste management practices, and market conditions, reflecting a complex economic landscape in the United States.[18,19]

2.6 Construction cost

Following is a comparison of alternative investments for energy and electricity production that might compete with WtE plants, which is summarised in Table 1. A coal power plant is similar to a WtE plant but the main fuel is coal. Building a coal power plant can cost from €1.66 million to €4.16 million per MW installed capacity. The cost mostly depends on the technology used within the plant, so called subcritical coal-fired power plants have efficacy of 30% and are the cheapest while the Ultra-supercritical have the highest efficiency at 45% and are on the expensive side [20]. Operation and maintenance costs depend on a lot of factors, but taking the US as example the cost is between \$20 and \$40 per MWh according to the U.S. Energy Information Administration [21].

On the other hand renewables are starting to be more competitive on the energy market. A solar farm with a capacity of 1 MW would cost between €822 000 and €1 million. Rooftop solar panels are far more expensive with €3,33 per watt [22]. Solar PV stood for 45% of total global electricity generation investments in 2022, which shows clearly in what direction the world is moving [23].

In 2019, the cost of onshore wind farms was €1.3 million per MW and €2.3 million for offshore wind farms. Wind farm investment costs have been declining for more than 10 years and the trend seems to be continuing, while operation costs stay the same. Number of projects is growing, especially the offshore farms since it is easier to transport the blades and turbines on ship. There seems to be potential for large offshore wind farms near coasts due to more constant winds [24].

Hydroelectric Power plants building costs on average €1.9 million per MW capacity. It is one of the most expensive electric generation technologies. Prices are steady but could increase due to higher labour and material costs in the future. Despite this, hydro power is one of the most important renewable energy sources. It is very reliable and production can be planned in advance by storage and dams [25].

Finally, the construction cost of an incineration plant with conventional technology is in the range of €6-10 million per MW. The price can be even higher for a plant with more advanced technology, for example advanced gasification brings up the price to €8-12 million per MW and pyrolysis can cost €7-11 million per MW [26].

Table 1. Construction cost of energy production units

Energy production unit	Cost range per MW in €
Coal power plant	1.6 million - 4.16 million
Solar farm	822 000 - 1 million
Wind farms onshore/offshore	1.3 million - 2.3 million
Hydroelectric power	1.9 million
Incineration plant	6 million - 10 million

3. Discussions and conclusions

Environmental advantages seem very clear from the study done. There are obvious resources from waste that are not utilised. Using WtE facilities can solve multiple problems such as treatment of waste that cannot be recycled or taken care of in a safe manner. It removes the need for landfill and all the drawbacks and risks that they include such as methane emissions and groundwater contamination. Moreover it doesn't change the ecosystem in the vicinity to the same extent and the air pollution isn't directly toxic to humans. Producing energy from WtE reduces use of fossil fuels. For the countries that have a more renewable energy production, WtE can still be beneficial compared to just incineration without any energy production.

The main drawback seems to be the CO_2 emissions. There is an effective way of flue gas collection, but not the same for CO_2 . Good news is that there are multiple projects on the way that seem to focus on this problem and have different methods of solving it. The Norwegian Statkraft WtE facility means to capture carbon and store it under the sea bed near Bergen while the Finish company Fortum wants to recycle the captured carbon and use it in new materials. Either way this could be a positive incentive for future investments in WtE.

Regarding the economic viabilities for the different countries it depends on several factors. Sweden, France and USA are examples of economies which gain profit from the use of this technology, proving its economical viability. Ghana is an example of a country which has the potential and the reasons to use WtE technology, still it faces problems which can be common for other developing countries, such as the lack of infrastructure.

A downside with WtE plants is the high construction cost. It is almost 4 times the hydroelectric plant, offshore wind farms and even coal power plants. This proves the point that renewable energy production can be cheaper and more environmentally friendly. On the other hand this calculation does not include the operation and management costs nor possible profits.

In conclusion, while waste-to-energy presents economic prospects such as job creation, renewable energy generation, and waste management solutions, its viability hinges on a complex interplay of technological advancement, waste composition, regulatory compliance, and societal attitudes toward waste management. Each country faces its unique set of challenges and opportunities in harnessing the economic potential of waste-to-energy initiatives.

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